An Internship Report

Of

Data Science with Python

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B-Tech (CSE)

By

Roll Number of Student: - 25SCS1003005554

Name of Student: KAJAL SINGH

Batch: 2026-27

CANDIDATE'S DECLARATION

I, KAJAL SINGH, do hereby declare that the internship report titled “DATA SCIENCE WITH PYTHON” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: -

Student Name: KAJAL SINGH

Roll No.: 25SCS1003005554

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me throughout my internship in Data Science with Python.

I am thankful to Beeskilled for providing me with the opportunity to participate in this online internship and gain practical knowledge in the field of Data Science. The internship helped me develop an understanding of important concepts such as data analysis, data visualization, machine learning, and working with datasets using Python.

I would also like to express my sincere gratitude to IILM University, Greater Noida, and the faculty members of the School of Computer Science and Engineering for their continuous support, guidance, and encouragement.

I am grateful to everyone who directly or indirectly contributed to the successful completion of my internship and its assignments. Finally, I would like to thank my parents for their constant support and encouragement throughout my academic journey

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Signature:

Student Name: KAJAL SINGH

Roll No.: 25SCS1003005554

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3. INTERNSHIP COMPLETION CERTIFICATE

Internship OfferLetter

INTERNSHIP OFFER LETTER

BEESKILLED

4.

Date: 17/07/2026

To,
Kajal Singh

Subject: Offer For Internship

Dear Kajal Singh ,

We are pleased to offer you an online internship with **Beeskilled** in the domain of Data Science with Python. The internship will commence from 17/07/2026 and end on 28/08/2026.

During this period, you will be required to work on assigned projects and complete the tasks within deadlines. This internship is aimed at enhancing your practical knowledge and hands-on experience.

We look forward to your positive response and hope this internship will contribute meaningfully to your career growth.

Best regards,

Skaur

Simranjeet Kaur
Founder & Program Head

AICTE & MSME REG.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the online internship undertaken with BeeSkilled in the domain of Data Science with Python. The internship was carried out as an online learning and practical training programme and focused on developing fundamental and practical knowledge in Data Science using Python.

The primary aim of this internship was to develop an understanding of the Data Science workflow and gain hands-on experience in working with real-world datasets. During the internship, I was introduced to important areas such as data analysis, data cleaning, handling missing values, statistical analysis, data visualization, and introductory machine learning.

The internship followed a structured week-wise learning approach that combined theoretical understanding with practical assignments and projects. During the first week, I worked with Python libraries such as Pandas and NumPy for creating and analysing Data Frames and handling missing data. In the following weeks, I performed data visualization using a COVID-19 dataset and learned introductory Machine Learning concepts using Linear Regression for predicting student performance.

The final stage of the internship involved a Data Analysis Project using the Titanic Dataset, where survival insights were explored through data analysis. The final project required the preparation of a Python notebook and project report.

Overall, the internship provided practical exposure to essential Data Science concepts and helped me improve my skills in Python-based data analysis, visualization, and basic Machine Learning.

4.2 Organization Profile

BeeSkilled is an online learning and internship platform that provides students with opportunities to develop practical knowledge and skills through structured learning programmes, assignments, projects, and assessments. The platform focuses on helping learners gain hands-on experience in different technical domains and apply their knowledge through practical tasks.

I completed my online internship through BeeSkilled in the domain of Data Science with Python. The internship programme provided a structured learning experience consisting of weekly assignments and practical projects. The activities focused on important areas of Data Science, including data analysis, data visualization, Machine Learning, and working with real-world datasets.

During the internship, I completed practical assignments using Python and relevant libraries and tools. The programme included working with data using Pandas and NumPy, performing data visualization, applying introductory Machine Learning techniques, and completing a final Data Analysis Project using the Titanic dataset.

The internship helped me strengthen my practical understanding of the Data Science workflow and develop skills through hands-on assignments. After completing the required internship activities and assessment, I received completion certificates for the relevant learning programmes.

4.3 Problem Statement

Data is generated in large quantities in different fields such as education, healthcare, business, and technology. However, raw data by itself may not provide useful information. To obtain meaningful insights, data needs to be collected, cleaned, analysed, and visualized properly.

One of the major challenges for students learning Data Science is gaining practical experience in applying theoretical concepts to real-world datasets. Students need hands-on knowledge of data handling, missing value treatment, statistical analysis, data visualization, and basic Machine Learning techniques.

The problem addressed through this internship was to develop practical skills in the Data Science workflow using Python. The internship focused on working with datasets, performing data analysis and visualization, and applying basic Machine Learning techniques.

Through practical assignments and projects, I gained experience in analysing different datasets, including Student Performance, COVID-19, and Titanic datasets.

4.4 Project Objectives

The main objectives of the Data Science with Python internship were:

- To understand the fundamental concepts and workflow of Data Science.
- To develop practical skills in Python for data analysis using libraries such as Pandas and NumPy.
- To learn how to create and manipulate Data Frames and handle missing values in datasets.
- To perform statistical analysis using measures such as mean, median, mode, minimum, and maximum values.
- To develop data visualization skills using tools such as Matplotlib and Seaborn.
- To perform Exploratory Data Analysis (EDA) using real-world datasets.
- To understand the basic concepts of Machine Learning and apply Linear Regression for prediction.
- To learn how to split data into training and testing datasets and evaluate model performance.
- To apply Data Science concepts in a practical project by analysing the Titanic dataset and identifying survival insights.

4.5 Scope of the Project

The scope of this internship was to develop practical knowledge and skills in the field of **Data Science using Python**. The internship covered important stages of the Data Science workflow, starting from working with datasets and handling missing values to data visualization and introductory Machine Learning.

The internship provided hands-on experience with different types of datasets. The work included creating and manipulating Data Frames, performing statistical analysis, handling missing values, and analysing datasets using Python libraries.

The scope also included **Exploratory Data Analysis and Data Visualization** using a COVID-19 dataset. Different visualizations, including trend lines, country comparisons, heatmaps, and scatter plots, were part of the assigned work.

Furthermore, the internship introduced basic **Machine Learning concepts** through a Student Performance prediction project using Linear Regression. The work involved training and testing data, making predictions, and evaluating model performance.

The final scope of the internship included applying the learned concepts to a practical **Titanic Dataset Analysis Project** to identify survival insights and prepare a Python notebook and project report.

Overall, the internship covered the fundamental practical areas of Data Science, including data analysis, visualization, basic Machine Learning, and project-based learning.

4.6 Technologies and Tools Used

The following technologies and tools were used during the internship:

Technology / Tool	**Purpose**
Python	Used as the primary programming language for Data Science tasks.
Pandas	Used for creating, manipulating, and analysing DataFrames and datasets.
NumPy	Used for numerical operations and handling numerical data.
Matplotlib	Used for creating data visualizations and graphical representations.
Seaborn	Used for statistical data visualization, including heatmaps and other plots.
Scikit-learn	Used for introductory Machine Learning and applying Linear Regression.

Jupyter Notebook / Google Colab	Used for writing and executing Python code and completing the project notebook.
CSV Datasets	Used as the source of data for analysis and Machine Learning activities.

The internship assignments specifically involved Pandas and NumPy, Matplotlib and Seaborn, and Scikit-Learn for Linear Regression.

4.7 System Architecture

The system architecture followed during the Data Science with Python internship represents the overall workflow used for processing and analysing data. The process begins with collecting and loading datasets into the Python environment. The data is then cleaned and processed to handle missing values and prepare it for analysis.

After preprocessing, Exploratory Data Analysis (EDA) and statistical analysis are performed to understand the dataset. Data visualization techniques are then used to identify patterns, trends, and relationships. For the Machine Learning activity, the processed data is divided into training and testing datasets, followed by model training, prediction, and evaluation.

The final stage involves interpreting the results and generating meaningful insights from the analysis.

Architecture Layers

- **Data Layer:** Datasets such as Student Performance, COVID-19, and Titanic datasets are used as input.
- **Data Processing Layer:** Python libraries such as Pandas and NumPy are used for data cleaning, handling missing values, and manipulating data.
- **Analysis Layer:** Statistical analysis and Exploratory Data Analysis are performed to understand the data.
- **Visualization Layer:** Matplotlib and Seaborn are used to represent data through graphs, plots, heatmaps, and other visualizations.
- **Machine Learning Layer:** Scikit-learn and Linear Regression are used for training, prediction, and model evaluation.
- **Results Layer:** The final results are interpreted to generate meaningful insights and conclusions.

FLOWCHART:

DATASETS
⬇
DATA LOADING
⬇
DATA CLEANING & PREPROCESSING
⬇
DATA ANALYSIS (EDA)
⬇
DATA VISUALIZATION
⬇
MACHINE LEARNING & PREDICTION
⬇
RESULTS & INSIGHTS

4.8 Methodology

The internship followed a structured **four-week methodology** consisting of practical assignments and project-based learning.

4.8.1 Week 1 – Data Analysis with Python

During the first week, I focused on understanding the basics of data handling and analysis using Python. I worked with **Pandas and NumPy** and created Data Frames manually.

The main activities included:

- Creating a Pandas DataFrame.
- Handling missing values using fillna().
- Sorting data based on a selected column.
- Calculating mean, median, and mode.
- Analysing a Student Marks dataset.
- Finding average, maximum, and minimum scores.

4.8.2 Week 2 – Data Visualization

During the second week, I worked on **Exploratory Data Analysis and Data Visualization** using a global COVID-19 dataset.

The objective was to practise data visualization using **Matplotlib and Seaborn**. The main activities included:

- Plotting trends of cases over time.
- Comparing the top five countries.
- Creating a heatmap.
- Creating a scatter plot.

This activity helped in understanding how graphical representations can be used to identify patterns and trends in data.

4.8.3 Week 3 – Machine Learning Introduction

During the third week, I was introduced to the basic concepts of **Machine Learning** through a Student Performance prediction project.

The project used student-related features such as:

- Hours studied
- Previous scores
- Sleep hours
- Sample question papers practiced

The main tasks included splitting the dataset into training and testing data, applying a **Linear Regression model**, predicting the performance index, and evaluating the model using measures such as **R^2 or MAE**.

4.8.4 Week 4 – Final Data Analysis Project

During the final week, I completed a **Data Analysis Project**. I selected the **Titanic Dataset** for the project.

The objective of the project was to analyse the dataset and identify **survival insights**. The required deliverables included:

- A Python notebook containing the project code.

- A project report.
- Submission of the completed project through the required link.

This final project allowed me to apply the concepts learned during the internship in a practical dataset analysis task.

4.9 Expected Outcomes

- Ability to perform data analysis using Python and work with datasets effectively.
- Practical understanding of Python libraries such as Pandas and NumPy for data manipulation and analysis.
- Ability to clean datasets and handle missing values.
- Knowledge of statistical analysis using measures such as mean, median, mode, minimum, and maximum values.
- Ability to perform Exploratory Data Analysis (EDA) to identify patterns and insights from data.
- Practical experience in creating data visualizations using Matplotlib and Seaborn.
- Understanding of basic Machine Learning concepts and the application of Linear Regression for prediction.
- Ability to divide data into training and testing datasets and evaluate model performance.
- Practical experience in analysing real-world datasets such as the COVID-19 and Titanic datasets.
- Ability to apply Data Science concepts in a complete practical project and generate meaningful insights.

4.10 Certificates of Completion and Communication Proof

During the **Data Science with Python internship at BeeSkilled**, I successfully completed the assigned learning activities, practical assignments, projects, and required assessments.

The internship provided practical exposure to different areas of Data Science, including **data analysis, data cleaning, data visualization, Exploratory Data Analysis, and introductory Machine Learning**. Throughout the internship, I completed weekly assignments and practical work using Python and real-world datasets.

After successfully completing the required internship activities and assessment, I received **completion certificates for Python and Data Science**. These certificates are included in this internship report as proof of successful completion of the respective programmes.

The practical assignments and project work completed during the internship have also been organized and uploaded to **Google Drive** for reference and verification. The submitted work includes relevant Python notebooks, assignments, dataset analysis, and the final Titanic Data Analysis Project.

Project Work and Assignment Proof

Google Drive Folder:

https://drive.google.com/drive/folders/1reJA727PKLexe1kd0qfn_4_tUt4vwkag

https://drive.google.com/drive/folders/1T471gISu2E4rekGGQh_ZNukt6jzjyBnu

https://drive.google.com/drive/folders/1V7ojmE3xw6i-JQ7472mk46TkLiYeZeGi

https://drive.google.com/drive/folders/1ajbZqpyrAnQZmd4gZS45WyGNI6HhY2bY

The folder contains:

- Week 1 – Data Analysis with Python
- Week 2 – Data Visualization
- Week 3 – Introduction to Machine Learning
- Week 4 – Titanic Data Analysis Project
- Python notebooks and assignment files
- Supporting project documents

Conclusion

The successful completion of this internship helped me gain practical knowledge and hands-on experience in **Python and Data Science**. The internship improved my understanding of the complete basic Data Science workflow, from working with raw data and performing analysis to visualization and applying introductory Machine Learning techniques.

5. BIBLIOGRAPHY/REFERENCES

- **BeeSkilled** – Internship and learning resources.
- **Python Documentation** – Official Python programming documentation.
- **Pandas Documentation** – Data analysis and manipulation resources.
- **NumPy Documentation** – Numerical computing resources.
- **Matplotlib Documentation** – Data visualization resources.
- **Scikit-learn Documentation** – Machine Learning resources.
- **Titanic Dataset** – Dataset used for the final Data Analysis Project.